DRUGS · ANTIBIOTICS: OTHER CLASSES

# Metronidazole

Metronidazole is a nitroimidazole that kills anaerobic bacteria and some protozoa. In head and neck practice it is the anaerobic partner added to a penicillin or cephalosporin for dental, peritonsillar, deep neck, otogenic and sinogenic intracranial infection, and for prophylaxis when the pharynx is opened. Its classic warnings are alcohol and warfarin.

## 1 CHAPTER ONE At a glance

The drug on one page.

### 1.1 Summary

METRONIDAZOLE

| | SUMMARY |
|---|---|
| **Class** | Nitroimidazole |
| **How it works** | Reduced inside anaerobic cells to reactive nitro radicals that break DNA; bactericidal |
| **Spectrum** | Obligate anaerobes (Bacteroides, Prevotella, Fusobacterium, Peptostreptococcus, Clostridia), protozoa (Trichomonas, Giardia, Entamoeba); no activity against aerobes or Actinomyces |
| **ENT uses** | With a beta-lactam for quinsy, deep neck infection, dental abscess, Lemierre syndrome, otogenic and sinogenic intracranial abscess, chronic and complicated sinusitis, necrotising ulcerative gingivitis; head and neck surgical prophylaxis; C. difficile (oral) in mild disease where vancomycin is not available |
| **Adult dose** | Oral 400 mg three times daily (or 500 mg three times daily); intravenous 500 mg every 8 hours; rectal 1 g every 8 hours |
| **Child dose** | 7.5 mg/kg three times daily orally or intravenously (maximum 500 mg) |
| **Avoid** | Alcohol during and for 48 hours after; caution in liver disease |

| | SUMMARY |
|---|---|
| **Pregnancy** | Avoid high-dose regimens; normal doses used when needed |

## 2 CHAPTER TWO How it works

Mechanism and handling.

### 2.1 Mechanism

A prodrug activated only in the low-redox environment of anaerobes, which is why it is selective. Resistance is rare.

### 2.2 Handling

Almost completely absorbed orally (so intravenous use is only for patients who cannot swallow), penetrates abscesses and the brain very well, metabolised in the liver.

## 3 CHAPTER THREE Using it in ENT

Indications and doses.

### 3.1 Mixed infections of the head and neck

Odontogenic, peritonsillar, parapharyngeal and retropharyngeal infections and **Lemierre syndrome** (Fusobacterium necrophorum thrombophlebitis of the internal jugular vein) are mixed or anaerobic: give a penicillin or cephalosporin with metronidazole, and drain collections.

### 3.2 Intracranial complications

For otogenic or sinogenic **brain abscess**, subdural empyema and venous sinus thrombosis, metronidazole with high-dose ceftriaxone is the standard empirical combination because of its excellent brain penetration.

### 3.3 Prophylaxis

Given with cefuroxime or co-amoxiclav at induction for clean-contaminated surgery such as laryngectomy and oral cavity resection.

## 4 CHAPTER FOUR Safety

What goes wrong.

### 4.1 Adverse effects

- Metallic taste, nausea, furred tongue, dark urine.
- **Disulfiram-like reaction** with alcohol: flushing, vomiting and tachycardia.
- **Peripheral and optic neuropathy** and encephalopathy with prolonged courses (beyond about 10 days or high totals); seizures.
- Hepatotoxicity, severe in Cockayne syndrome.

### 4.2 Interactions

**Warfarin** (marked rise in INR), alcohol, lithium (toxicity), phenytoin and busulfan; phenobarbital lowers its levels.

## 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

## 5.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- Selectivity for anaerobes and why that is.
- Organisms of **Lemierre syndrome** and its treatment.
- The **alcohol** and **warfarin** interactions; neuropathy with long courses.

## 5.2 Common pitfalls

**COMMON PITFALLS**

- Giving intravenous metronidazole to a patient who can swallow.
- Forgetting to warn about alcohol.
- Long courses without review for neuropathy.

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